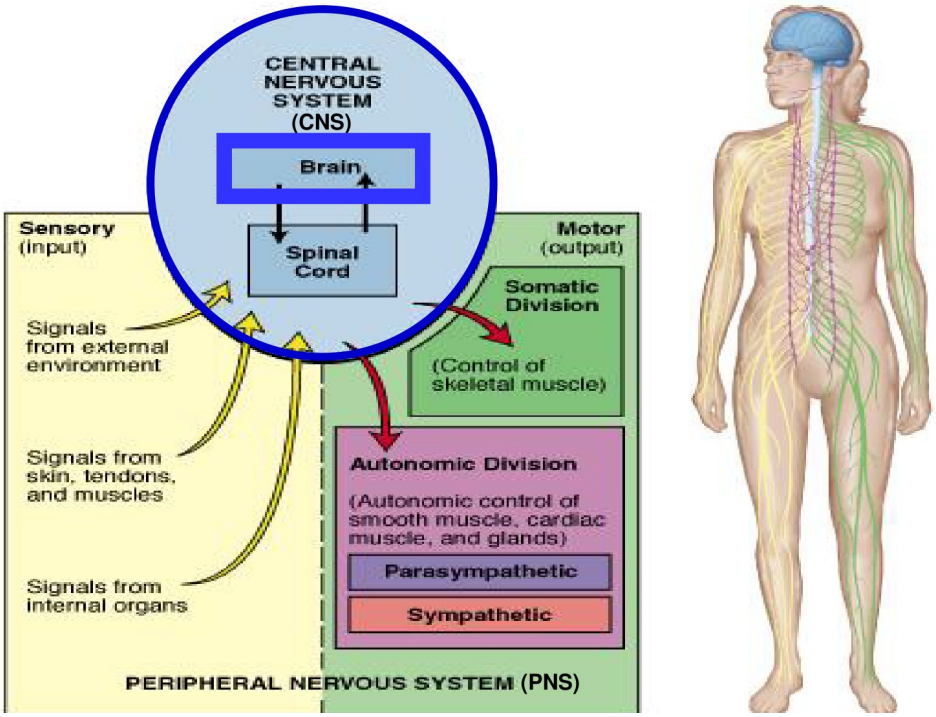
01 Neural organisation: input, integration & output

Central idea: sensory information is integrated by cortical and subcortical networks to support perception, voluntary action, arousal, cognition and motivated behaviour.

• The **central nervous system (CNS)** consists of the brain and spinal cord. The **peripheral nervous system (PNS)** carries sensory information toward the CNS and motor commands away from it.



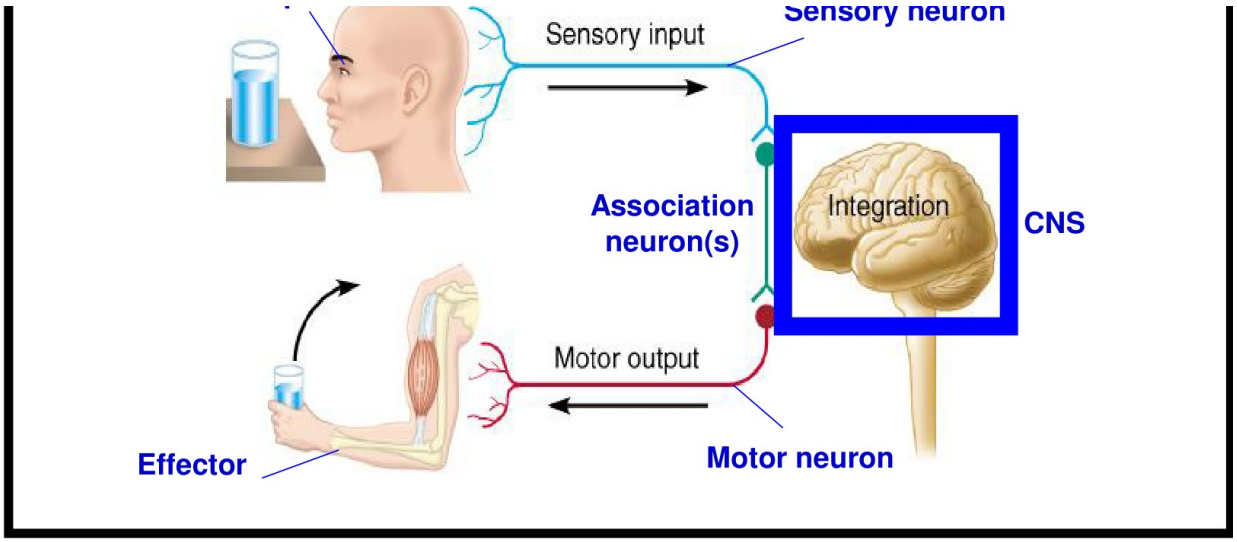
Sensory input and somatic / autonomic output

Division	Information or target
Sensory / afferent	External environment; skin, tendons and muscles; internal organs.
Somatic motor	Skeletal muscle; voluntary movement is a major function.
Autonomic motor	Smooth muscle, cardiac muscle and glands; sympathetic and parasympathetic divisions.

• Sympathetic activity commonly mobilises energy; parasympathetic activity commonly supports maintenance and conservation. These are broad functional tendencies.

02 Neural pathways: input to response

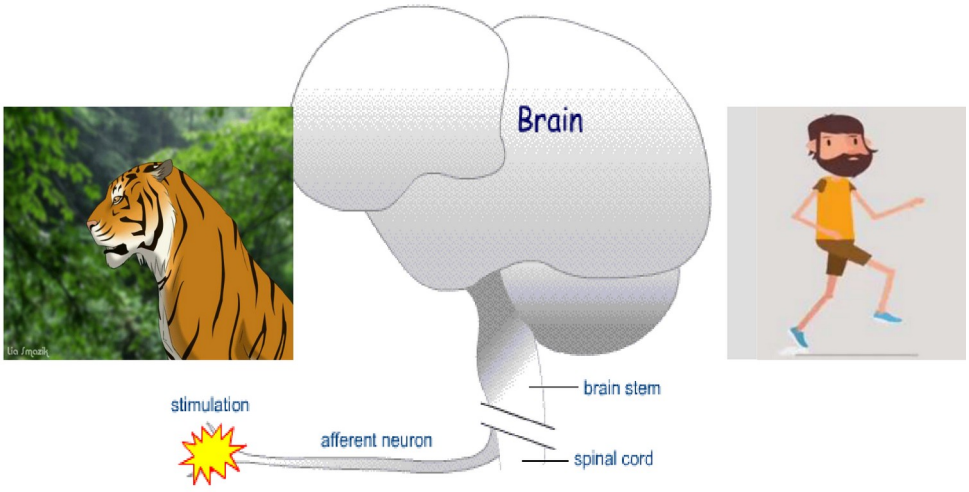
Stimulus → **receptor** → sensory neuron → CNS integration by association neurons → motor neuron → **effector** → response



Receptor → sensory input → integration → motor output

- **Sensory input:** detection of a stimulus by receptors. **Integration:** interpretation of sensory input within the brain and spinal cord. **Motor output:** activation of an effector, such as muscle.



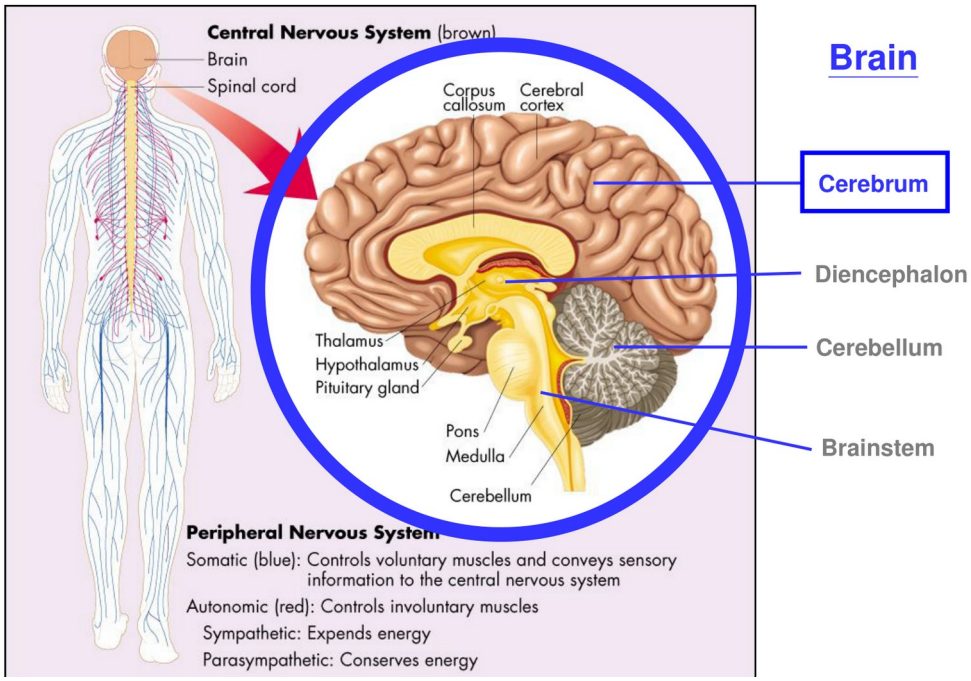


Environmental stimulus → brain processing → behaviour

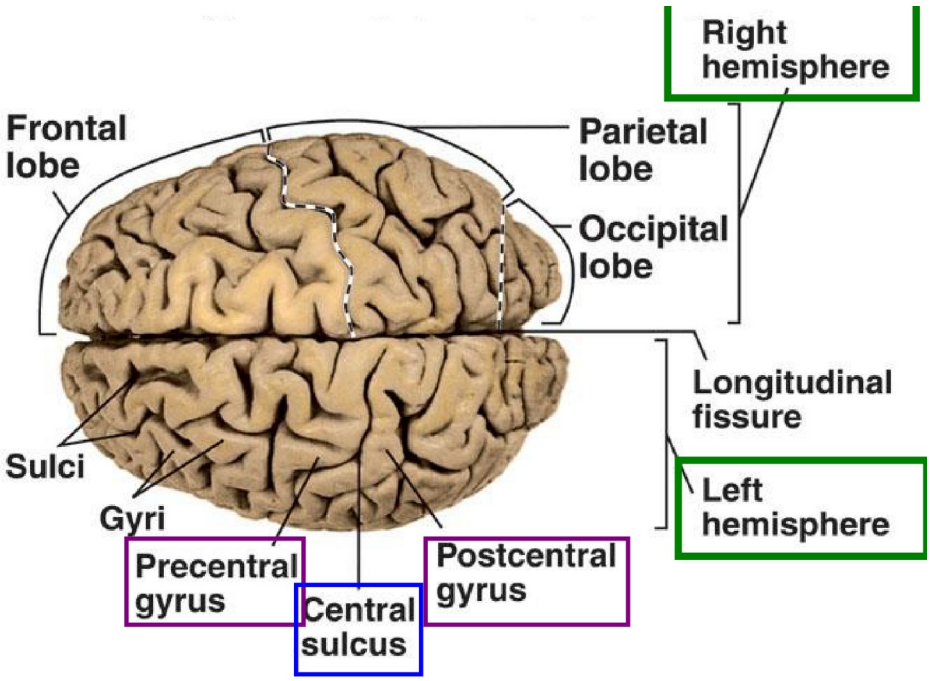
- An association neuron is an **interneuron**. Integration can involve several interneurons and multiple CNS regions; it is not necessarily a single synapse.

03 Cerebrum and anatomical landmarks

• The brain includes the **cerebrum, diencephalon, brainstem and cerebellum**. The diencephalon includes the thalamus and hypothalamus; the brainstem includes midbrain, pons and medulla.



Brain regions and sagittal landmarks

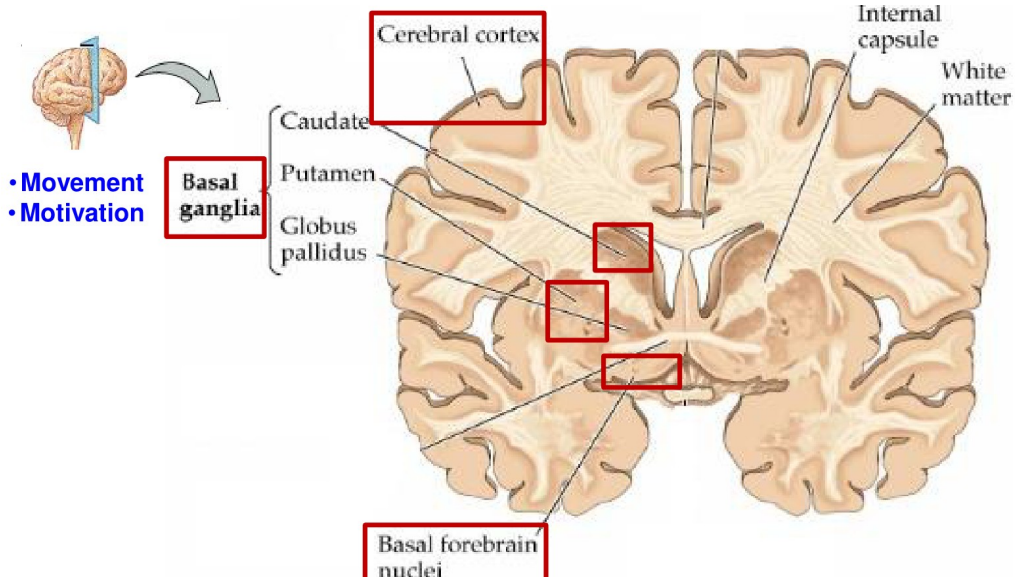


Cerebral hemispheres, sulci and gyri

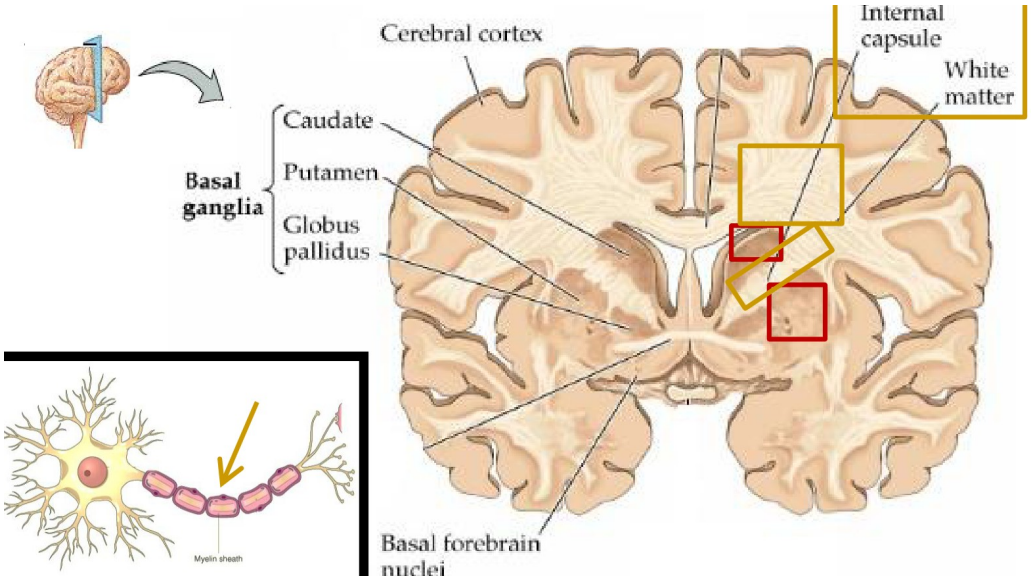
• **Gyri** are cortical ridges; **sulci** are grooves. The **cerebral cortex** is the outermost layer of neural tissue of the cerebrum.

04 Gray matter, white matter and nuclei

Structure	Composition / role
Gray matter	Mainly neuronal cell bodies, with dendrites and synapses. The cerebral cortex is gray matter.
White matter	Predominantly myelinated axons connecting neural regions. The internal capsule is a major white-matter pathway.
Basal ganglia (basal nuclei)	Deep gray-matter nuclei involved in movement and motivation. The illustrated nuclei include caudate, putamen and globus pallidus.



Gray matter: cortex and deep nuclei

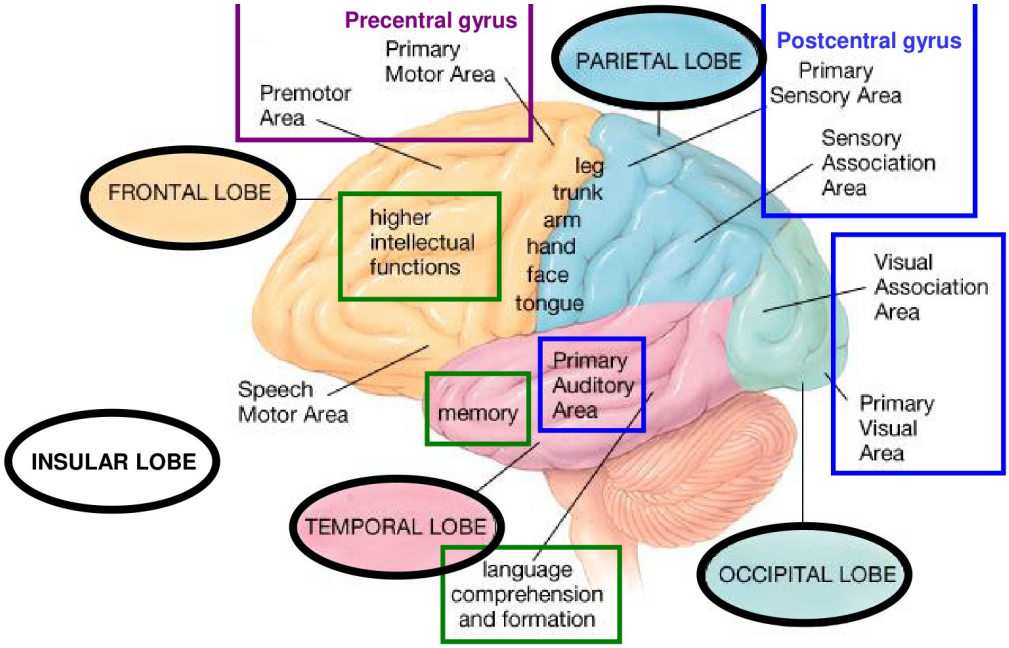


White matter: internal capsule and myelinated connections

- **Deep gray matter is not cortex.** White matter carries connections between cell populations; the basal ganglia should not be confused with the nearby basal forebrain nuclei.

05 Cerebral lobes: a functional map

- The cerebral lobes collectively support **sensation, communication, memory, understanding and voluntary movement**.

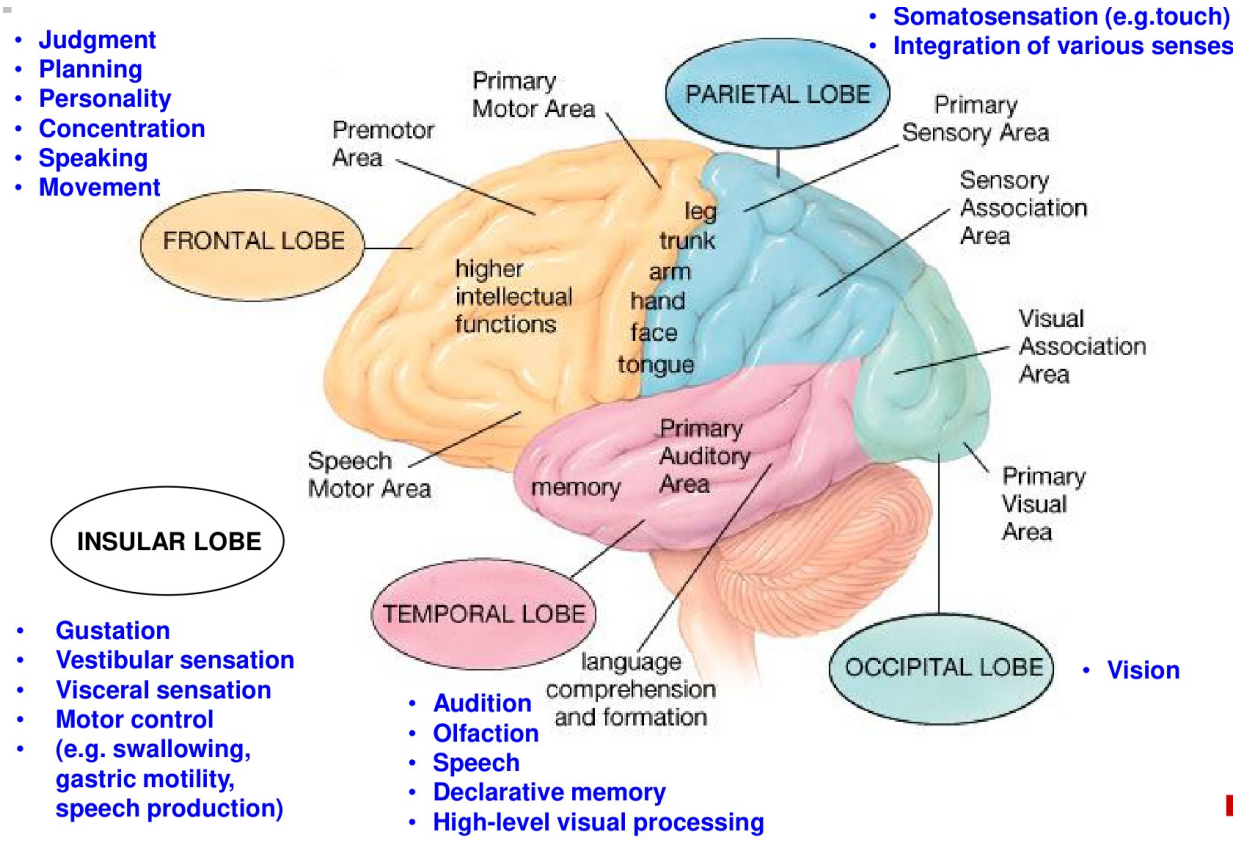


Cerebral lobes and cortical landmarks

Lobe	Principal functions
Frontal	Judgment, planning, personality, concentration, speaking and movement.
Parietal	Somatosensation, such as touch; integration of information from different senses.
Occipital	Vision.
Temporal	Audition, olfaction, speech, declarative memory and high-level visual processing.
Insular	Gustation, vestibular and visceral sensation, and motor control; examples include swallowing, gastric motility and speech production.

- **Declarative memory** concerns consciously accessible facts and events. High-level visual processing supports recognition and interpretation, beyond initial visual detection.

06 Lobes and distributed cortical functions



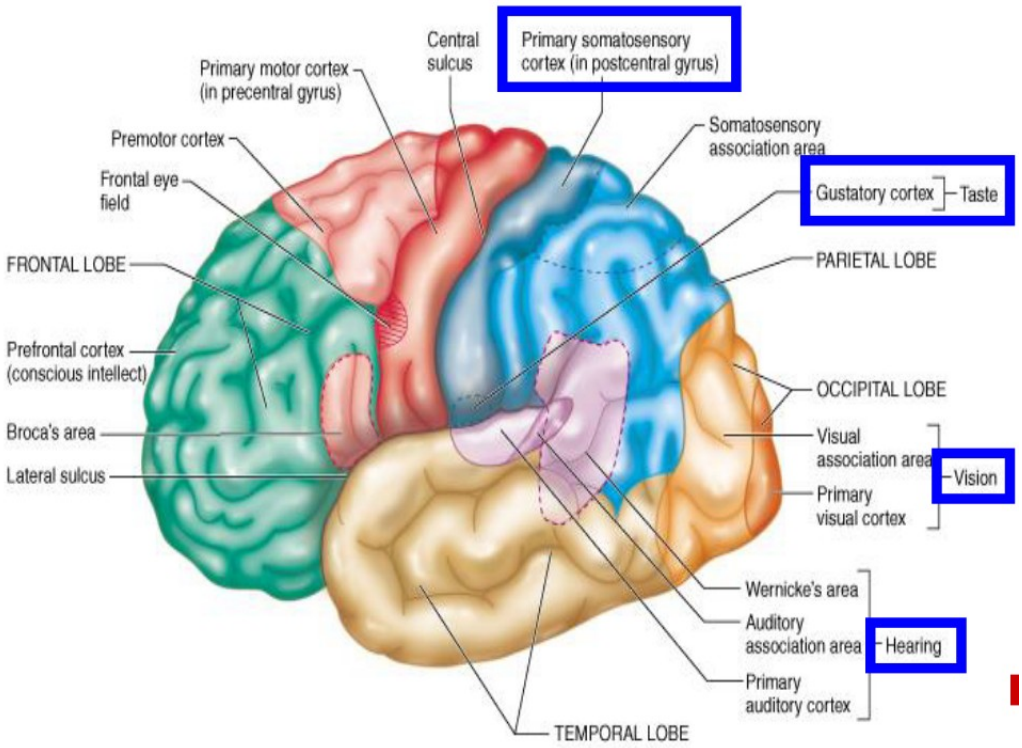
Cerebral lobes: locations and functions

Functional category	Main contribution
Sensory areas	Conscious awareness of sensation.
Motor areas	Control of voluntary movement.
Association areas	Integration of diverse information and higher mental functions, including learning.

- A lobe contains several functional areas. A complex function such as speech or memory depends on **interacting regions**, rather than one isolated lobe.
- The insula lies deep to the lateral sulcus and is not exposed on a simple lateral surface view. Taste and vestibular functions also involve adjacent or distributed cortical regions.

07 Sensory cortex: location and meaning

Sensory area	Location / contribution
Primary somatosensory cortex	Postcentral gyrus, parietal lobe; conscious somatic sensation.
Visual cortex	Occipital lobe; vision.
Auditory cortex	Temporal lobe; hearing.
Olfactory cortex	Includes temporal-lobe regions; smell.
Gustatory cortex	Insula and adjacent opercular cortex; taste.
Vestibular cortex	A distributed network including insular and parietal regions; perception of head motion and orientation.



Primary sensory cortex and association areas

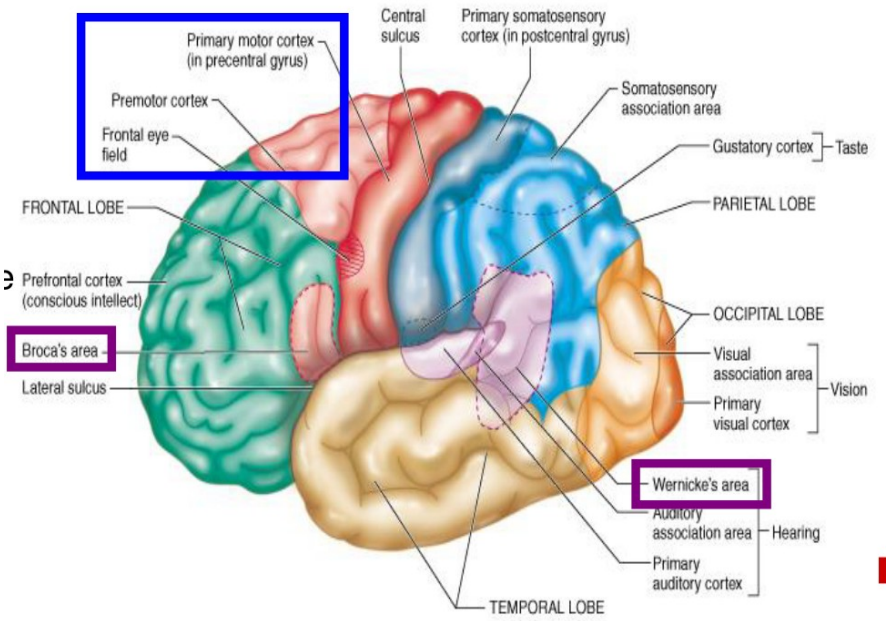
Sensory input → primary sensory processing → association processing → recognition and interpretation

- **Detection and recognition differ.** Feeling contact is distinct from identifying an object by its shape or texture. Hearing a sound is distinct from understanding a word.

08 Motor cortex and speech

Motor cortex organises voluntary action.
Broca's area supports speech production; **Wernicke's area** supports comprehension.

Frontal motor area	Role
Primary motor cortex	Precentral gyrus; contributes descending commands for voluntary movement.
Premotor cortex	Preparation and organisation of movements.
Frontal eye field	Control of voluntary eye movements.
Broca's area	Motor aspects of speech: articulation and coordination of movements needed to produce speech.



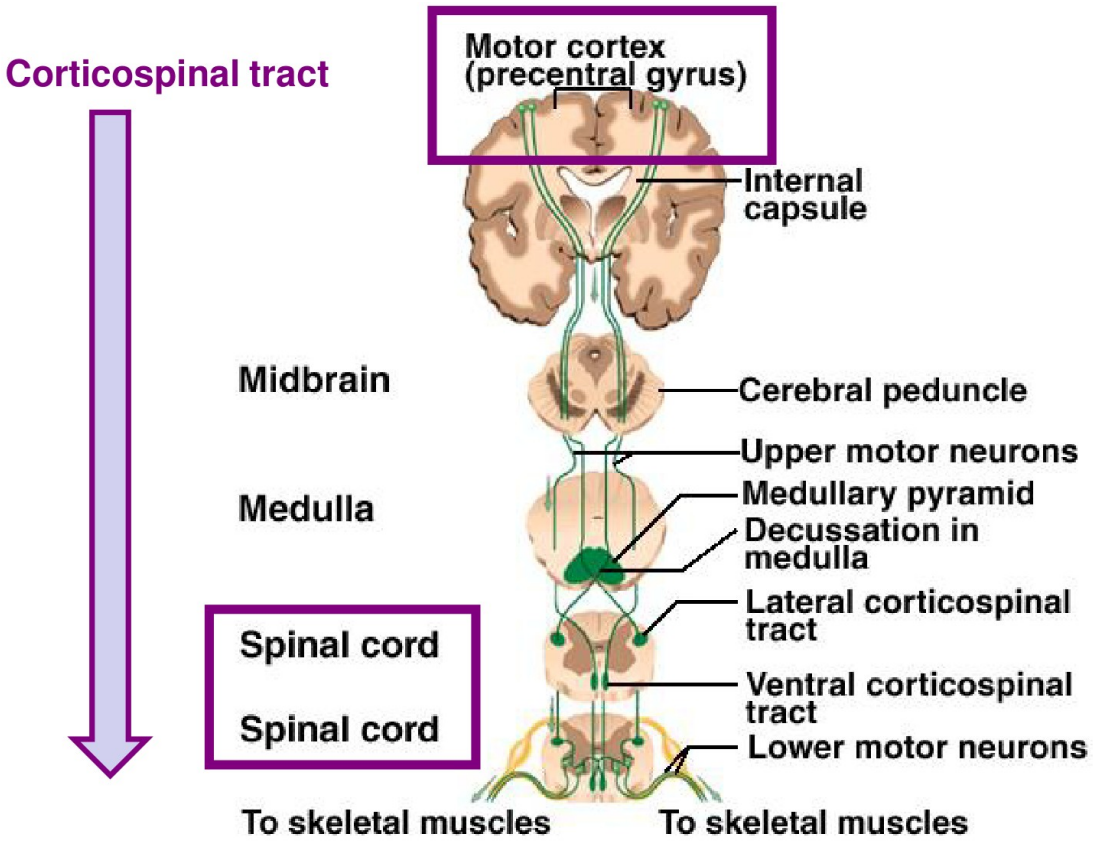
Motor cortex, frontal eye field and language areas

Language component	Main contribution
Broca's area: motor speech	Organises speech production and the movements needed for articulation.
Wernicke's area: sensory speech	Links recognisable spoken words to meaning; supports speech comprehension and selection of appropriate words for expression.

- "Sensory speech" refers to **language comprehension**, not merely hearing. Wernicke-related processing interprets word meanings and translates words into thoughts.

09 Corticospinal tract and crossing

Motor cortex → internal capsule → cerebral peduncle in midbrain → medullary pyramid → **pyramidal decussation** → lateral corticospinal tract → spinal motor circuitry → skeletal muscle

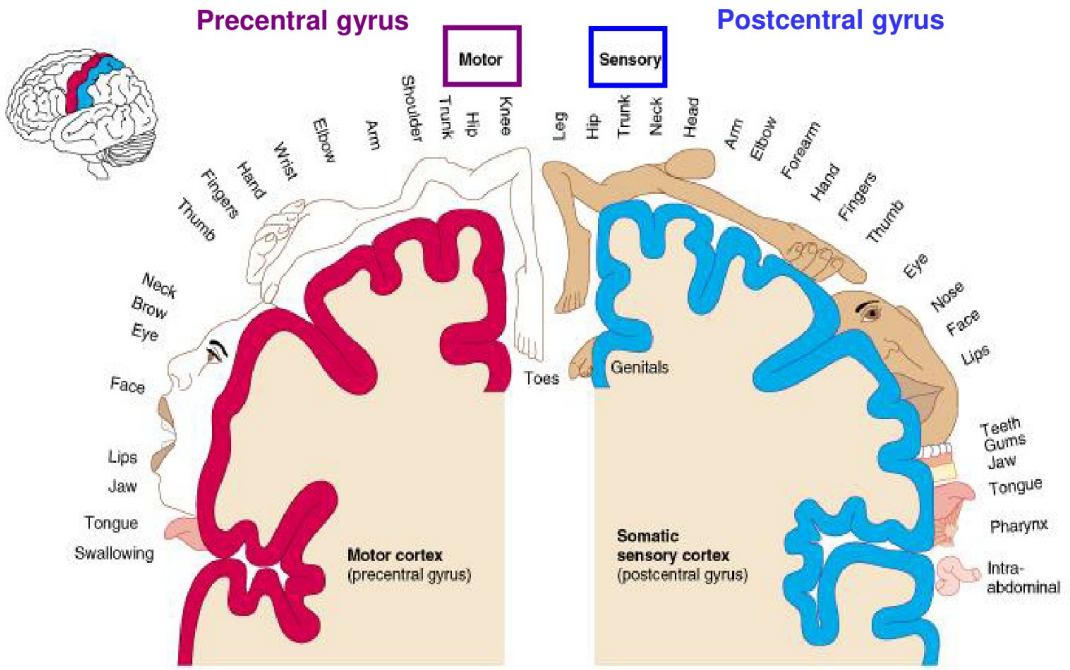


Corticospinal pathways: upper and lower motor neurons

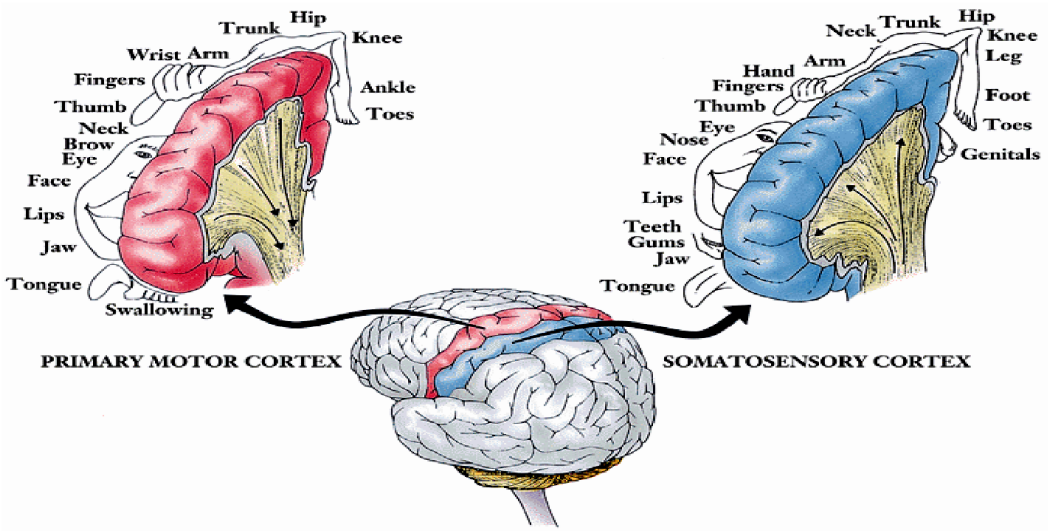
- **Upper motor neurons** descend from the cortex. They influence **lower motor neurons** in the spinal cord, often through interneurons; lower motor neuron axons reach skeletal muscle.
- Most corticospinal fibres **cross in the caudal medulla** and enter the lateral corticospinal tract. This explains much of the contralateral control of the limbs.
- The anterior (ventral) corticospinal component initially descends without this crossing; many fibres cross near their spinal termination and influence axial muscles bilaterally.
- Consequently, a lesion above the main decussation tends to cause contralateral limb weakness; a lateral tract lesion below it tends to cause ipsilateral weakness below the lesion.

10 Cortical homunculi

• A **cortical homunculus** is a topographic representation of body regions in the cortex. There are sensory and motor maps.



Motor map in the precentral gyrus; sensory map in the postcentral gyrus.



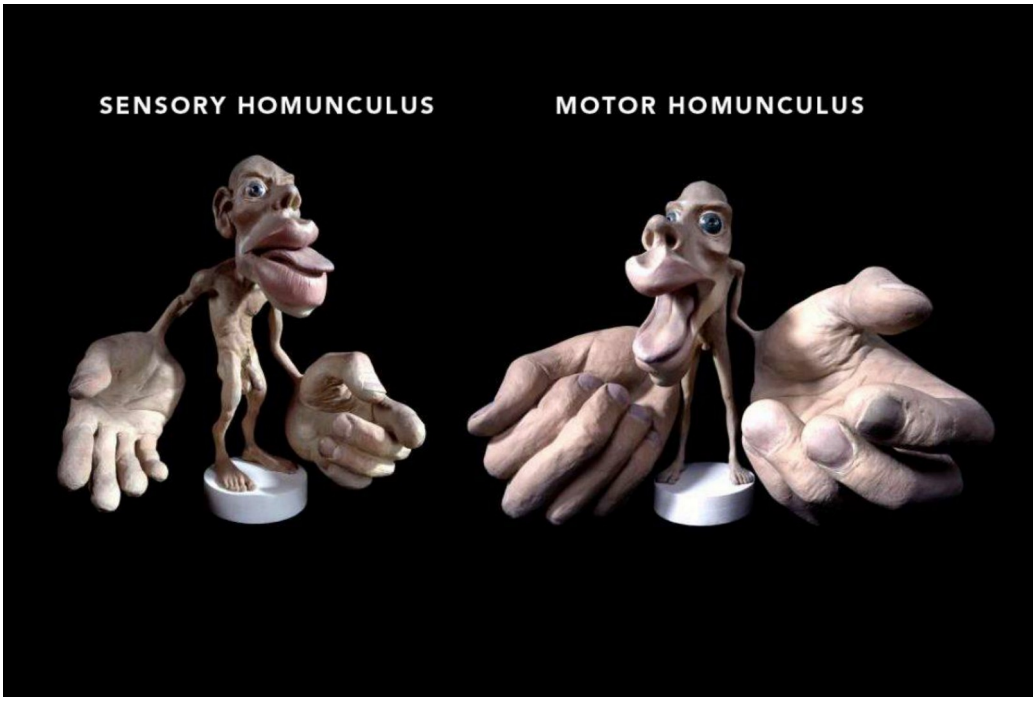
Adjacent cortical representations show disproportionately large hands, face and tongue.

• The leg and foot are represented mainly medially; the upper limb lies more superior-laterally; face and tongue lie more inferolaterally. The two maps are related but are not identical.

Cortical representation reflects processing demand, not physical body-part size.

11 Why the homunculus is distorted

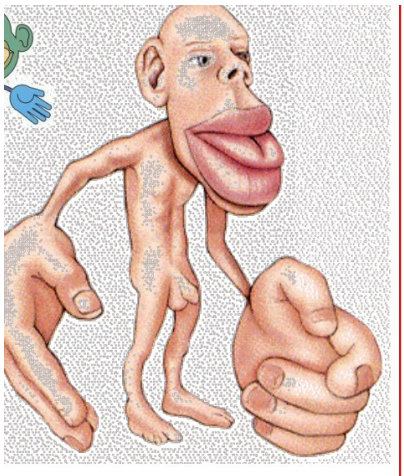
Greater sensory resolution / finer motor control → **greater cortical representation**.
Body-part size alone does not determine map size.



Sensory and motor homunculi

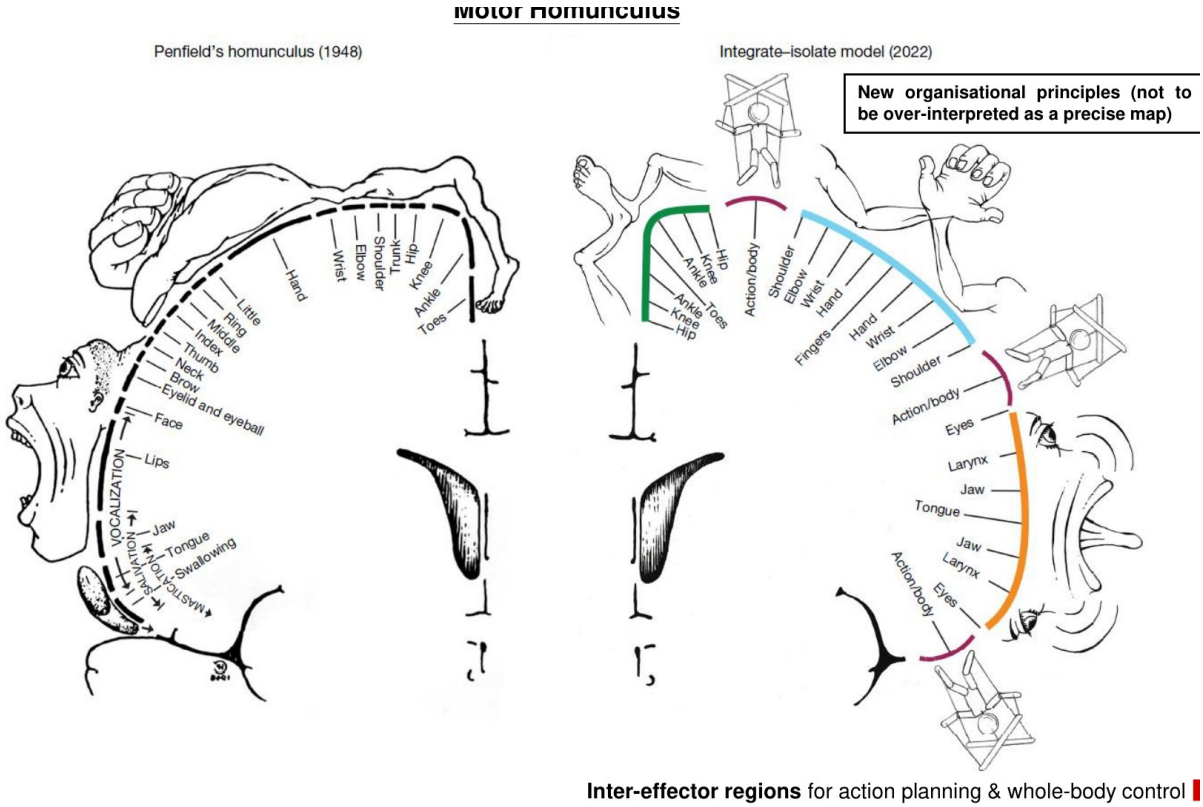
Classroom question: why is representation disproportionate?

- Sensory representation is related to **sensory receptor density and discriminative resolution**. A densely sampled region needs more processing capacity.
- Motor representation is related to **fine control and the complexity of independent movements**. The teaching shorthand “density of motor neurons” points toward innervation demands, but is not a literal universal scaling law.
- Large hands and lips in these models therefore indicate specialised sensory or motor processing. They do not mean that those body parts are physically largest.



Disproportionate cortical representation

12 Beyond a single motor strip map



Classical motor map and inter-effector organisation

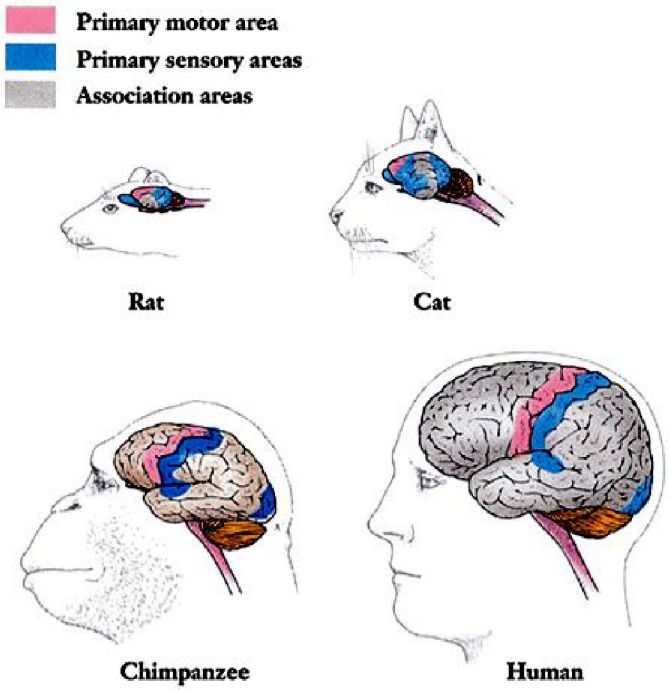
- The classical homunculus is a useful **orientation map**, rather than a perfectly precise one-to-one assignment of each cortical point to one muscle.
- The newer organisational model includes **inter-effector regions** between effector-related representations. These regions contribute to action planning and whole-body control.

> Effector-specific control + inter-effector integration → coordinated actions involving the whole body

- Read the new diagram as an organisational principle. Do not over-interpret it as an exact anatomical map or assume that motor control is a simple continuous list of isolated body parts.

13 Association cortex and higher functions

• Association areas are not primarily responsible for initial sensory reception or direct motor output. They integrate information for **higher mental functions**, including learning.



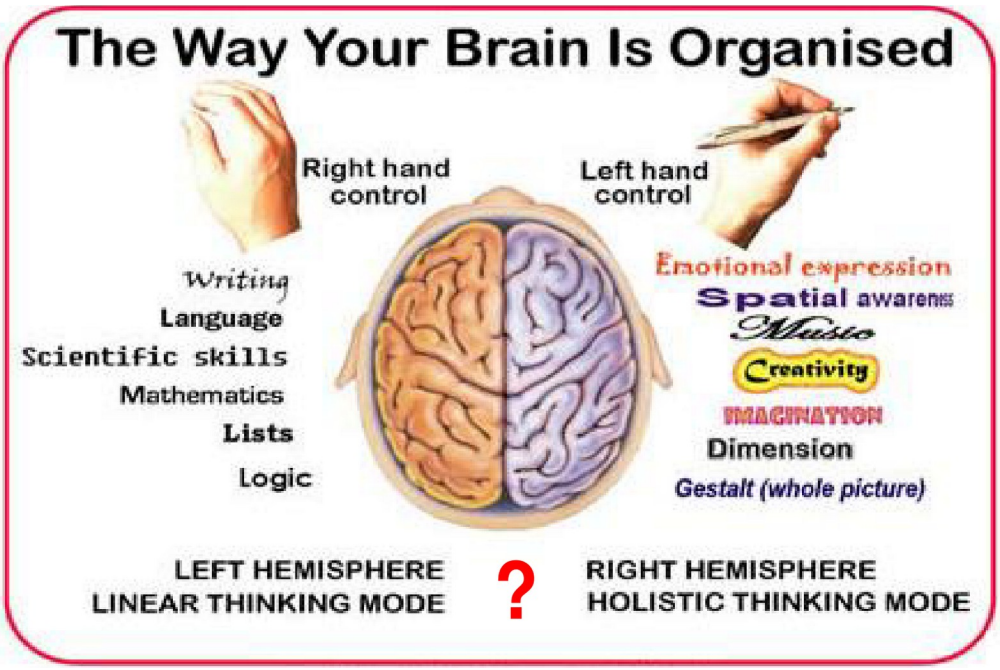
Primary and association cortex across species

Association area	Processing contribution
Somatosensory	Identification of shape, texture and orientation.
Visual	Recognition and evaluation of what is seen.
Auditory	Differentiation of sound as speech, music or noise.
Wernicke's	Interpretation of word meanings; translation of words into thoughts.
Integrative	Formation of thoughts from sensory inputs.

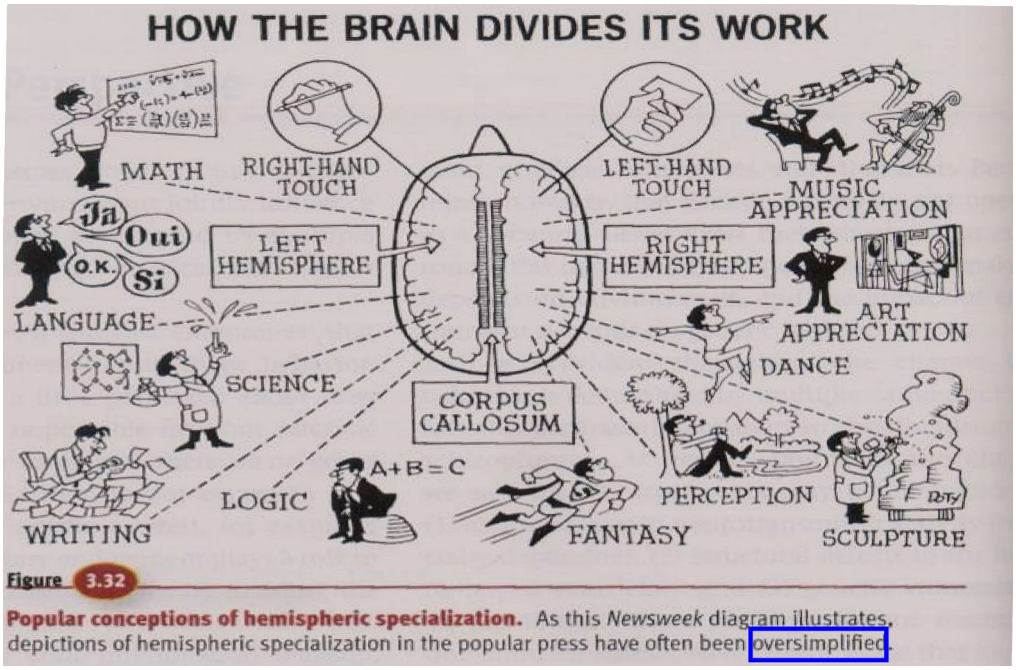
Sensory features → integrated representation → recognition / meaning → thought and appropriate action

• The comparative illustration emphasises the extensive association cortex in humans; complex cognition still depends on coordinated networks.

14 Hemispheric specialisation



Popular hemispheric assignments: interpret with caution

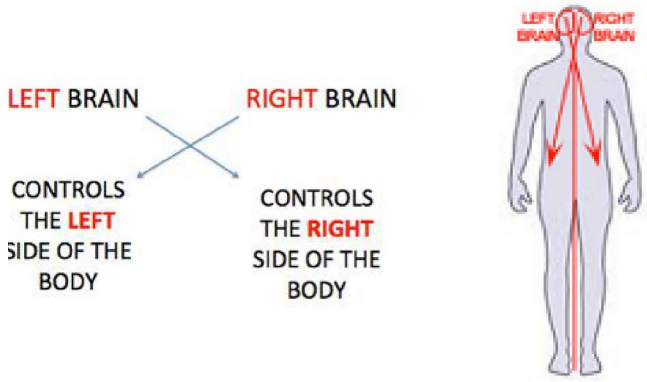


Hemispheric specialisation is often oversimplified

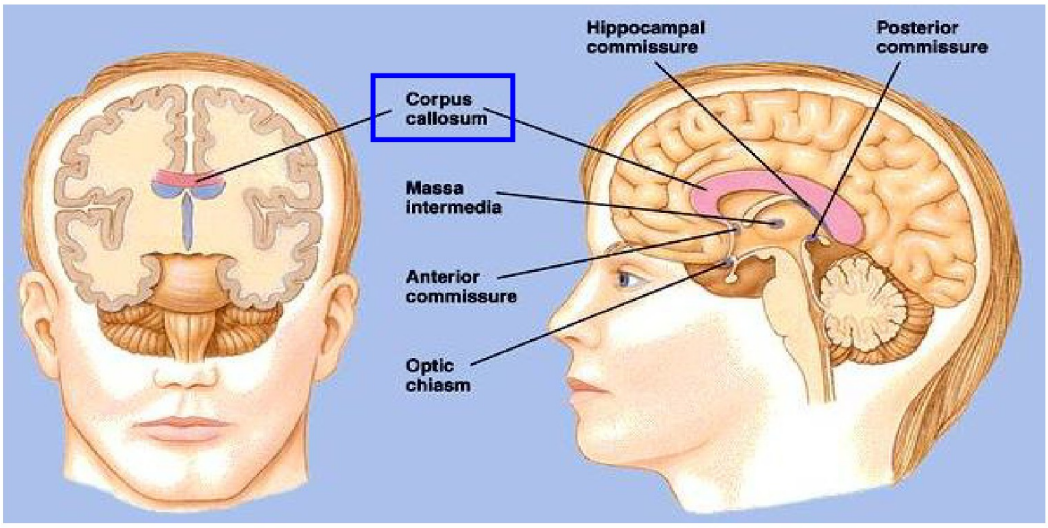
- **Lateralisation means relative specialisation, not exclusive ownership.** It does not divide people into a logical "left-brained" type and a creative "right-brained" type.

15 Lateralisation and interhemispheric links

General tendency	Interpretation
Left hemisphere	Language is left-dominant in most individuals; mathematical and logical processing often shows left-sided specialisation.
Right hemisphere	Often contributes strongly to visuospatial processing and aspects of emotion, music and artistic processing.
Both hemispheres	Work together; complex skills cannot be assigned exclusively to one side.



Crossed control: left hemisphere mainly controls the right body; right hemisphere mainly controls the left body.

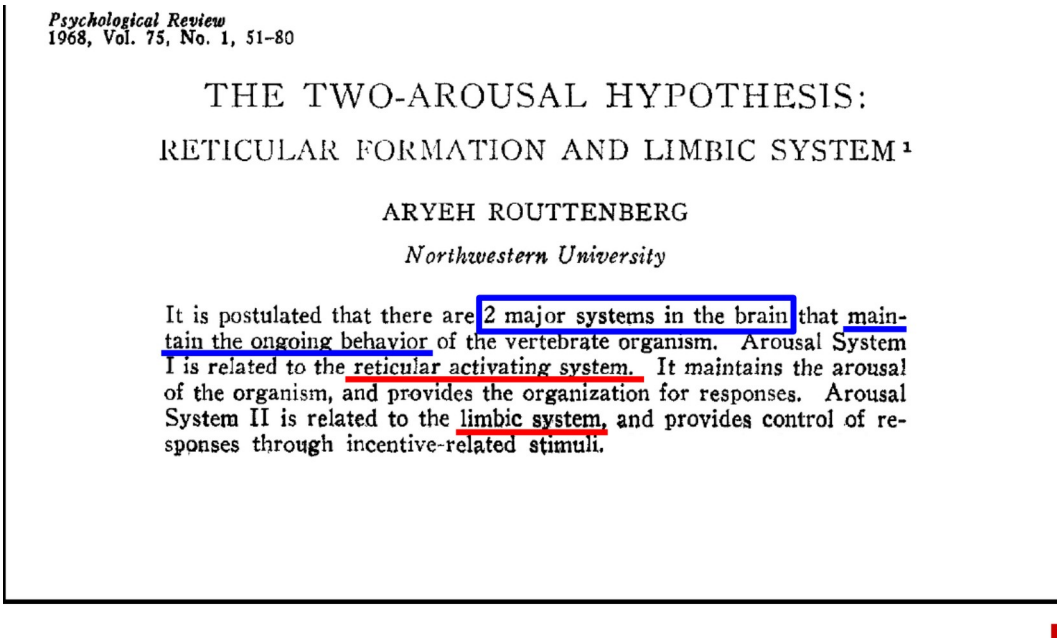


The corpus callosum is the main connection between corresponding cortical regions across hemispheres.

- The **corpus callosum** is a commissural white-matter bundle. The figure also labels anterior, posterior and hippocampal commissures, the optic chiasm and massa intermedia.
- Do not confuse these structures: the optic chiasm carries crossing visual fibres; the massa intermedia is an interthalamic adhesion, not the main cortical commissure.

16 Arousal and the limbic system

Reticular formation + limbic networks → an aroused brain with behaviour shaped by motivation and significance



Psychological Review
1968, Vol. 75, No. 1, 51–80

THE TWO-AROUSAL HYPOTHESIS:

RETICULAR FORMATION AND LIMBIC SYSTEM[1]

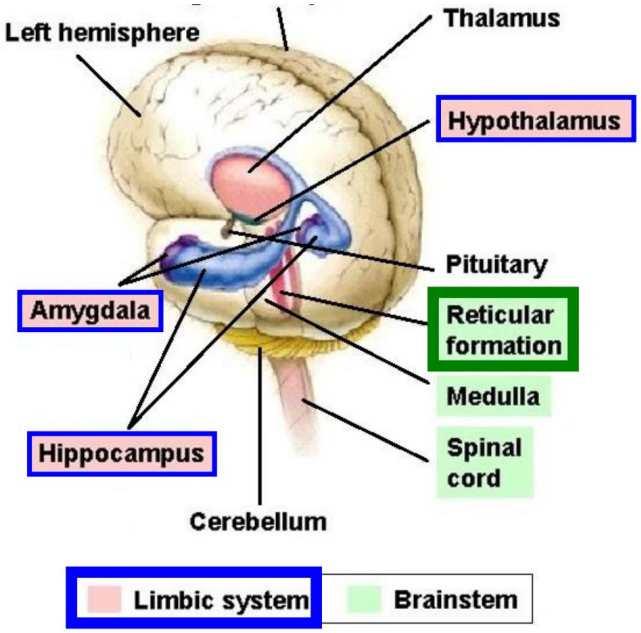
ARYEH ROUTTENBERG

Northwestern University

It is postulated that there are 2 major systems in the brain that maintain the ongoing behavior of the vertebrate organism. Arousal System I is related to the reticular activating system. It maintains the arousal of the organism, and provides the organization for responses. Arousal System II is related to the limbic system, and provides control of responses through incentive-related stimuli.

The two-arousal hypothesis (Aryeh Routtenberg, 1968): arousal and incentive-related control of responses.

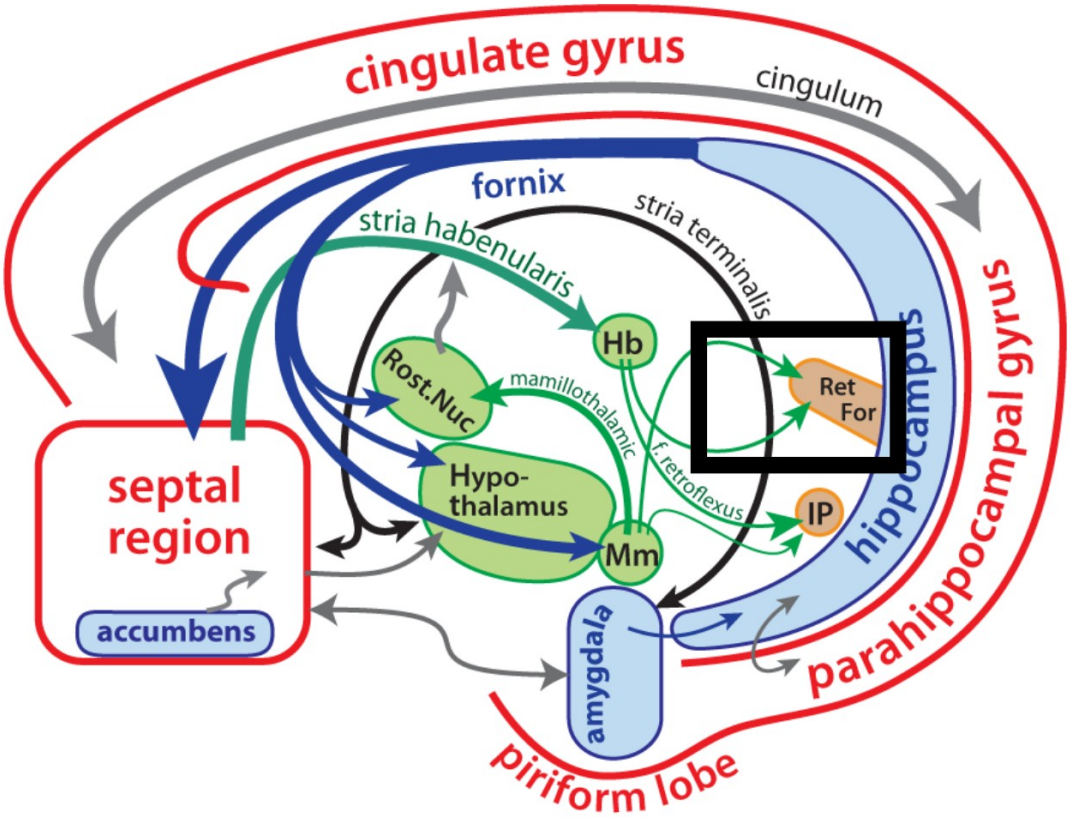
- In this historical framework, one system is related to the reticular activating system: it maintains arousal and organises responses. The other is related to the limbic system: it influences responses to incentive-related stimuli.



Anatomical context for interacting reticular and limbic systems, including thalamus, hypothalamus, amygdala and hippocampus.

- These systems **function together**. Arousal allows responsiveness, while emotional, memory and motivational context helps determine what matters and how to respond.

17 Limbic connections: reading the network



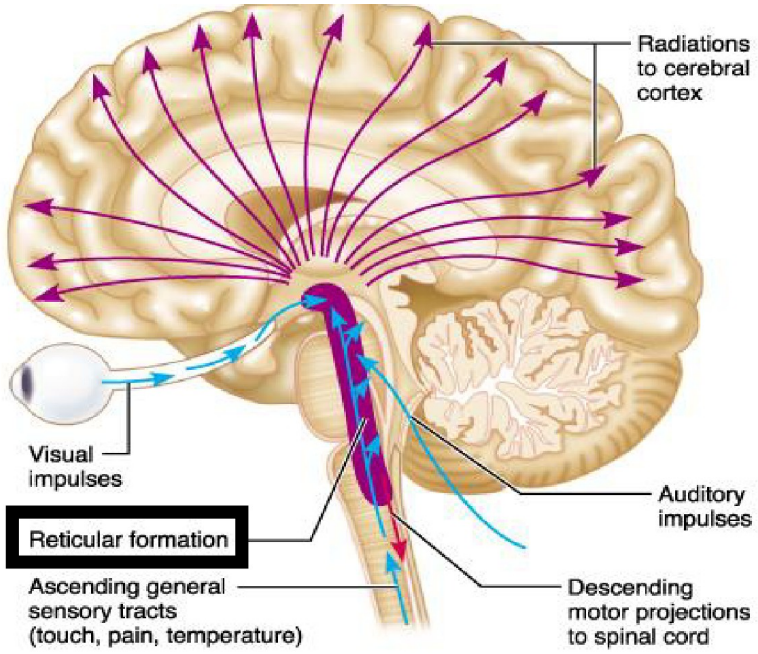
Interconnected cortical and subcortical limbic structures; the reticular formation connection is boxed.

Structures / tracts	How to read the diagram
Cortical and limbic regions	Cingulate gyrus, parahippocampal gyrus, piriform lobe, hippocampus, amygdala and septal region; the accumbens is also labelled.
Connecting tracts	Cingulum, fornix, stria terminalis, stria habenularis, mammillothalamic tract and tractus retroflexus.
Subcortical labels	Hypothalamus; mammillary bodies (Mm); habenula (Hb); interpeduncular nucleus (IP); rostral nuclear label (Rost. Nuc.); reticular formation (Ret. For.).

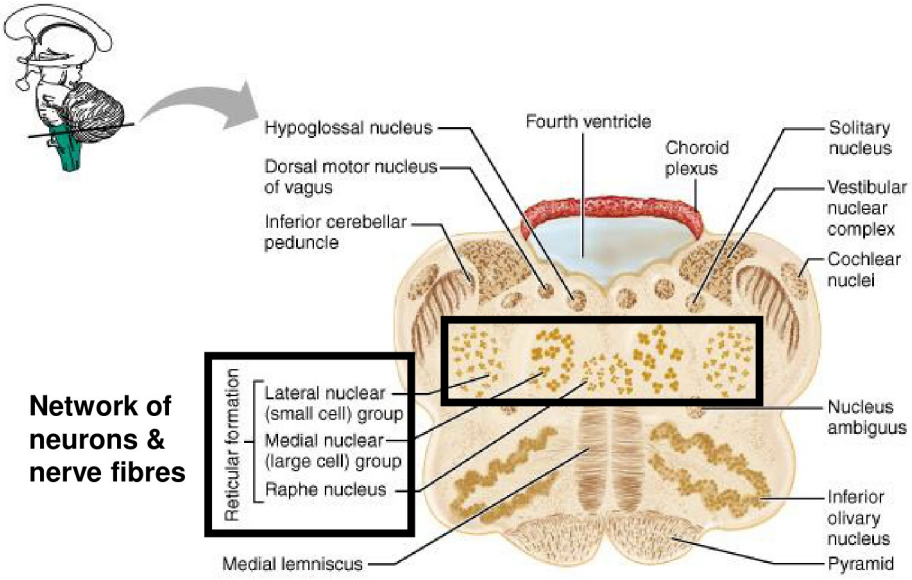
- Follow the arrowheads to identify direction. The multiple connections indicate network interaction; they should not be compressed into one linear pathway or all labelled as excitatory.
- The figure emphasises connectivity rather than neurotransmitter sign. A connection alone does not establish **positive feedback**.

18 Reticular formation: core organisation

• The **reticular formation (RF)** is a network of neurons and nerve fibres running through the **central core of the brainstem**.



Visual, auditory and ascending general sensory inputs converge; outputs include broad cortical projections and descending spinal projections.

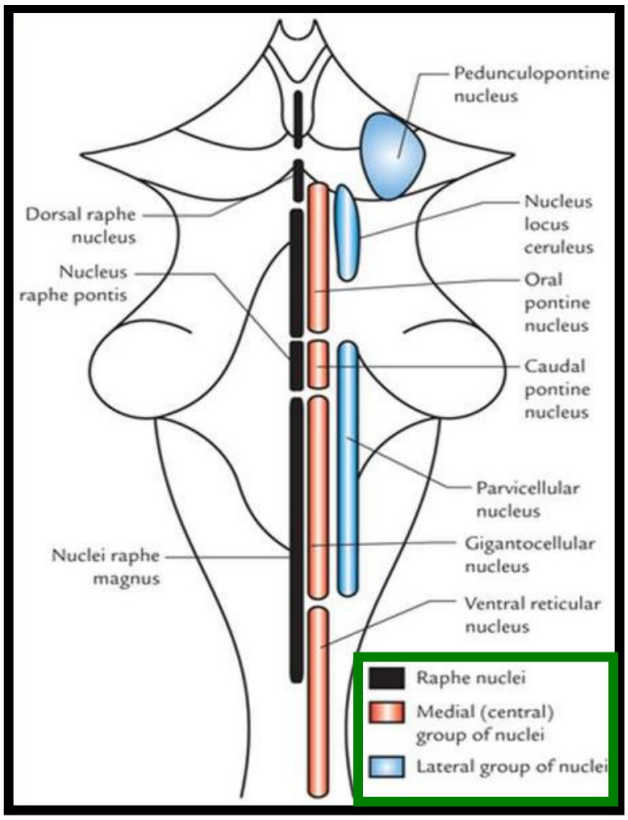


Reticular formation within the brainstem core

• Three major groupings are distinguished: **lateral nuclear group (small cells), medial nuclear group (large cells), and midline raphe nuclei**.

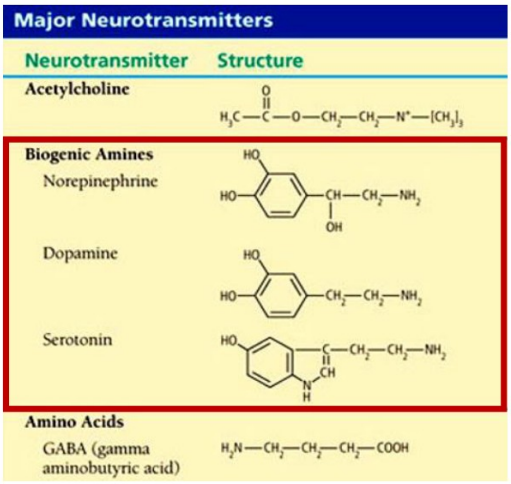
• The surrounding landmarks include hypoglossal, dorsal motor vagal, solitary, vestibular, cochlear and ambiguus nuclei, plus inferior olive, pyramid, medial lemniscus and inferior cerebellar peduncle.

19 Reticular nuclei and chemical diversity



Raphe, medial and lateral nuclear groups

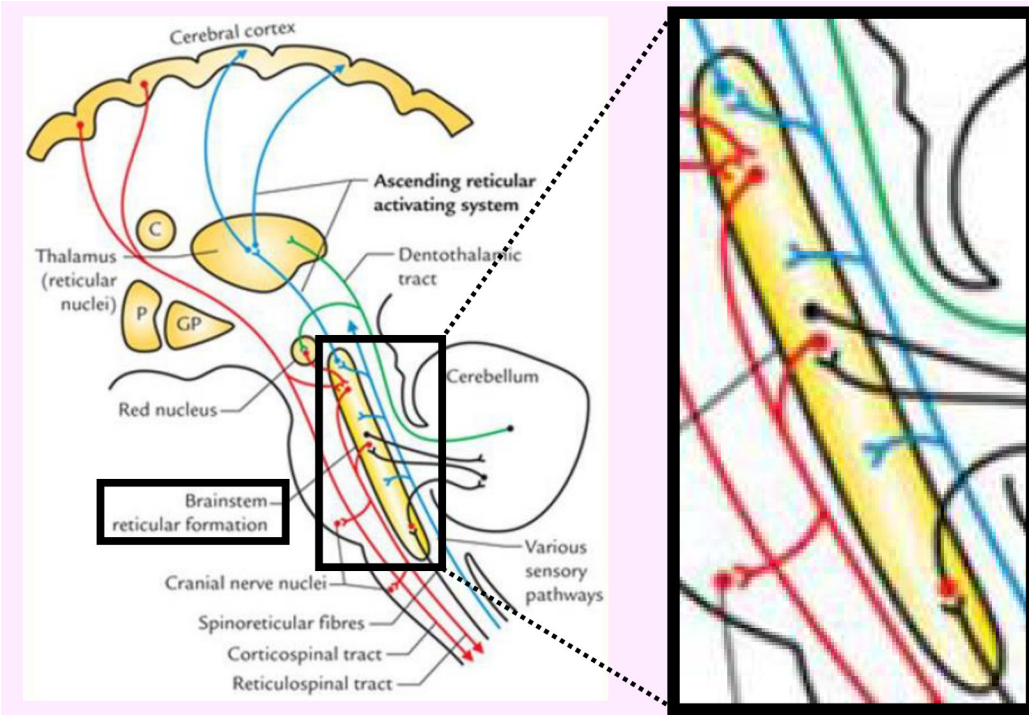
- The map labels dorsal raphe, nucleus raphe pontis, raphe magnus, oral and caudal pontine nuclei, parvicellular, gigantocellular and ventral reticular nuclei, locus coeruleus and pedunculopontine nucleus.
- **Monoaminergic neurons** make an important contribution to RF-related neural activity and behavioural state. Norepinephrine, dopamine and serotonin are the monoamines highlighted in the neurotransmitter chart.



Biogenic amines and other transmitter classes

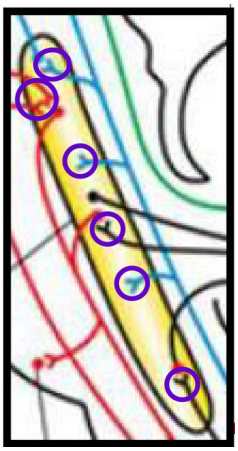
- **Acetylcholine is not a monoamine.** Gamma-aminobutyric acid (GABA) is also not a monoamine. Chemical diversity supports different modulatory roles; RF neurons are not all the same transmitter type.

20 Heterogeneous input, diffuse output

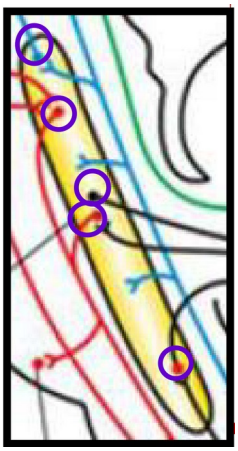


Converging inputs and ascending / descending connections

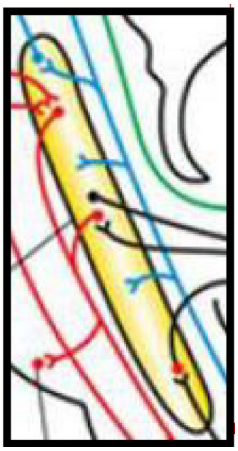
- RF receives a **continuous stream of multimodal sensory information**. Its inputs are heterogeneous: several modalities and pathways influence the network.
- Its outputs are **widespread and diffuse**. Changes in incoming activity alter outgoing activity and thereby change behavioural responsiveness.



Diverse inputs



Diffuse outputs

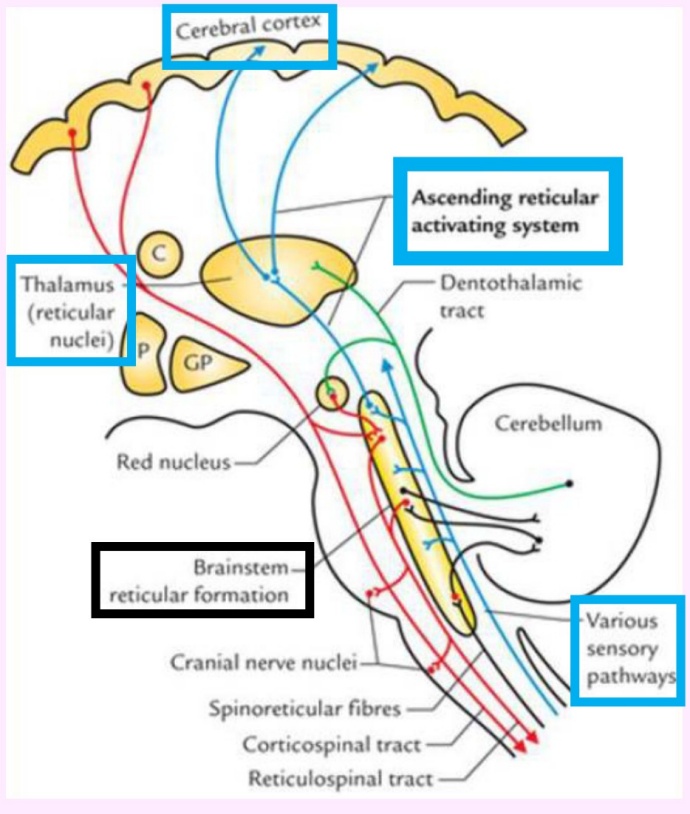


Network detail

Diverse inputs + interconnected neurons + chemical diversity → varied outputs and functions

21 Ascending reticular activating system

• The **ascending reticular activating system (ARAS)** includes RF nuclei with ascending projections that modulate the thalamus and cerebral cortex. It supports consciousness, alertness, attentiveness and awareness.



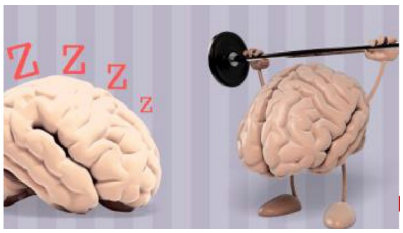
Ascending reticular activating system

Visual / acoustic / somatosensory impulses → RF and ascending arousal pathways → thalamic and cortical modulation → **alertness and attention**

• RF activation helps maintain normal wakefulness and awareness of environmental stimuli. It also influences the **sensory threshold**, affecting which inputs gain effective access to conscious processing.

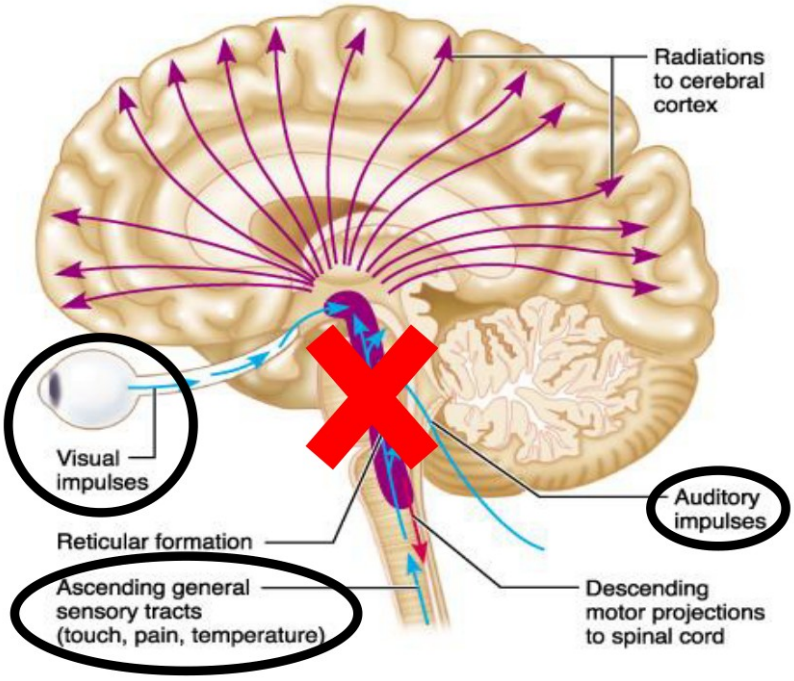
• RF interacts with **orexin neurons in the lateral hypothalamus** to regulate sleep and wakefulness. Orexin helps stabilise wakefulness.

• The ascending arousal system has multiple routes; it should not be treated as a single obligatory relay through the thalamic reticular nucleus. Thalamic arousal relays and broader forebrain pathways contribute.



Sleep and wakefulness are regulated brain states.

22 Sleep, reduced arousal and coma



Disruption of ascending cortical activation

Condition	Mechanism / consequence
Strong sensory event	Bright light or a ringing alarm → stimulation of arousal pathways → increased alertness or awakening.
Reduced sensory stimulation	Removal of visual, auditory and somatosensory input may favour drowsiness and sleep.
Sleep	Reduced activity in several wake-promoting pathways contributes to sleep, within an actively regulated sleep–wake system.
RF / ARAS injury	Disrupted cortical activation → reduced consciousness; substantial injury may produce coma.
Severe brainstem injury	Can be fatal, especially when vital respiratory and cardiovascular systems are also affected.
General anaesthetics	Can alter arousal-system function and consciousness; their effects are not limited to one RF site.

- **Coma** is prolonged unconsciousness in which even strong external stimulation cannot produce arousal. Normal sleep remains reversible by an appropriate stimulus.
- Reduced ARAS activity is a useful simplification for sleep onset, not a claim that the whole system switches off throughout sleep. Severe ARAS damage does not inevitably imply death by itself.

23 Habituation and selective responsiveness

Repeated, insignificant stimulus → declining response to that stimulus → attention remains available for **important or changing stimuli**

- **Habituation** is a form of learning in which the brain increasingly ignores repetitive, meaningless input while remaining sensitive to other stimuli.
- Example: a person may sleep through familiar traffic noise but awaken promptly to an alarm. Repetition and behavioural significance matter, rather than loudness alone.

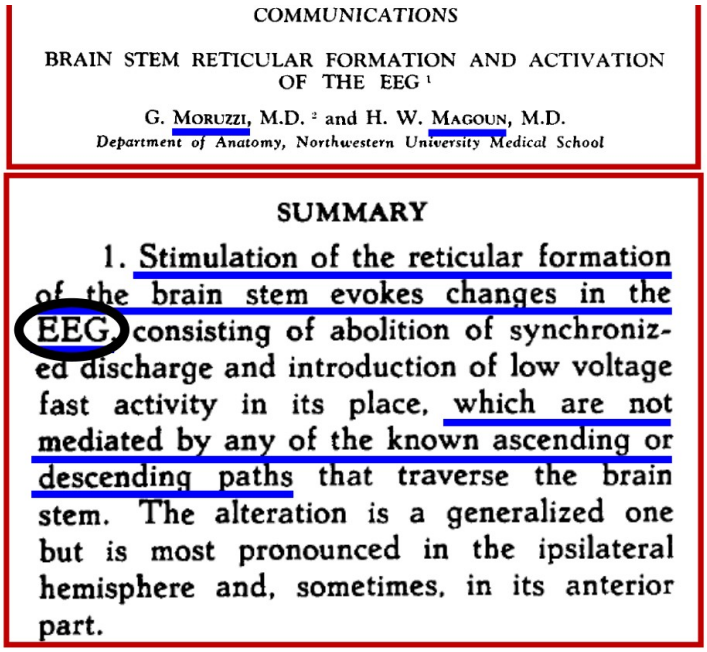


Habituation to a repeated background stimulus

Concept	Key distinction
Habituation	A stimulus-specific reduction in behavioural response with repetition.
General reduction in arousal	A broader change in responsiveness, such as drowsiness.
Coma	A pathological inability to be aroused even by powerful stimuli.

- Habituation does not mean that all sensory receptors stop working or that every incoming signal is blocked. The brain can still respond to a novel or meaningful event.

24 Evidence linking RF to cortical activation



COMMUNICATIONS

BRAIN STEM RETICULAR FORMATION AND ACTIVATION OF THE EEG [1]

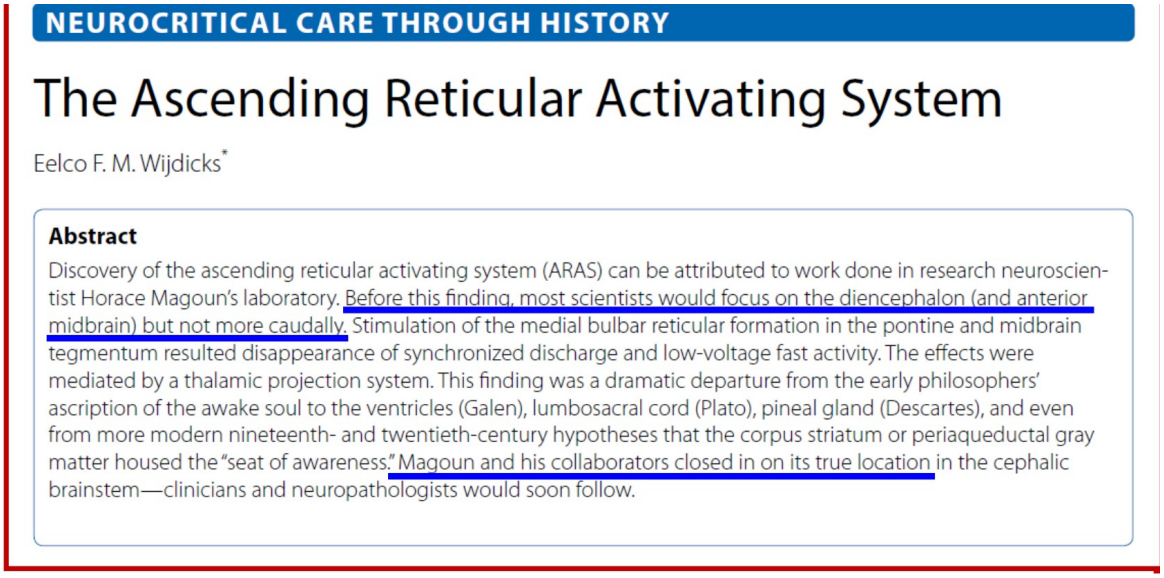
G. MORUZZI, M.D. [2] and H. W. MAGOUN, M.D.
Department of Anatomy, Northwestern University Medical School

SUMMARY

1. Stimulation of the reticular formation of the brain stem evokes changes in the EEG, consisting of abolition of synchronized discharge and introduction of low voltage fast activity in its place, which are not mediated by any of the known ascending or descending paths that traverse the brain stem. The alteration is a generalized one but is most pronounced in the ipsilateral hemisphere and, sometimes, in its anterior part.

Moruzzi and Magoun (1949): brainstem RF stimulation changes the electroencephalogram.

- Stimulation of the brainstem RF changed the **electroencephalogram (EEG)** from synchronised activity toward **low-voltage, fast activity**. The reported change was generalised, often strongest ipsilaterally and sometimes anteriorly.
- The historical interpretation was that this activation was not simply mediated by the known ascending or descending tracts passing through the brainstem.



NEUROCRITICAL CARE THROUGH HISTORY

The Ascending Reticular Activating System

Eelco F. M. Wijdicks*

Abstract

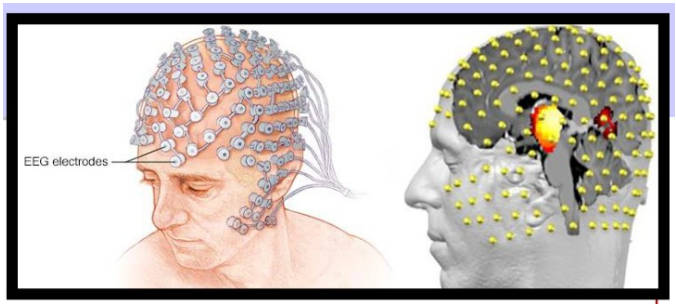
Discovery of the ascending reticular activating system (ARAS) can be attributed to work done in research neuroscientist Horace Magoun's laboratory. Before this finding, most scientists would focus on the diencephalon (and anterior midbrain) but not more caudally. Stimulation of the medial bulbar reticular formation in the pontine and midbrain tegmentum resulted disappearance of synchronized discharge and low-voltage fast activity. The effects were mediated by a thalamic projection system. This finding was a dramatic departure from the early philosophers' ascription of the awake soul to the ventricles (Galen), lumbosacral cord (Plato), pineal gland (Descartes), and even from more modern nineteenth- and twentieth-century hypotheses that the corpus striatum or periaqueductal gray matter housed the "seat of awareness." Magoun and his collaborators closed in on its true location in the cephalic brainstem—clinicians and neuropathologists would soon follow.

Historical perspective: ascending arousal systems (2019)

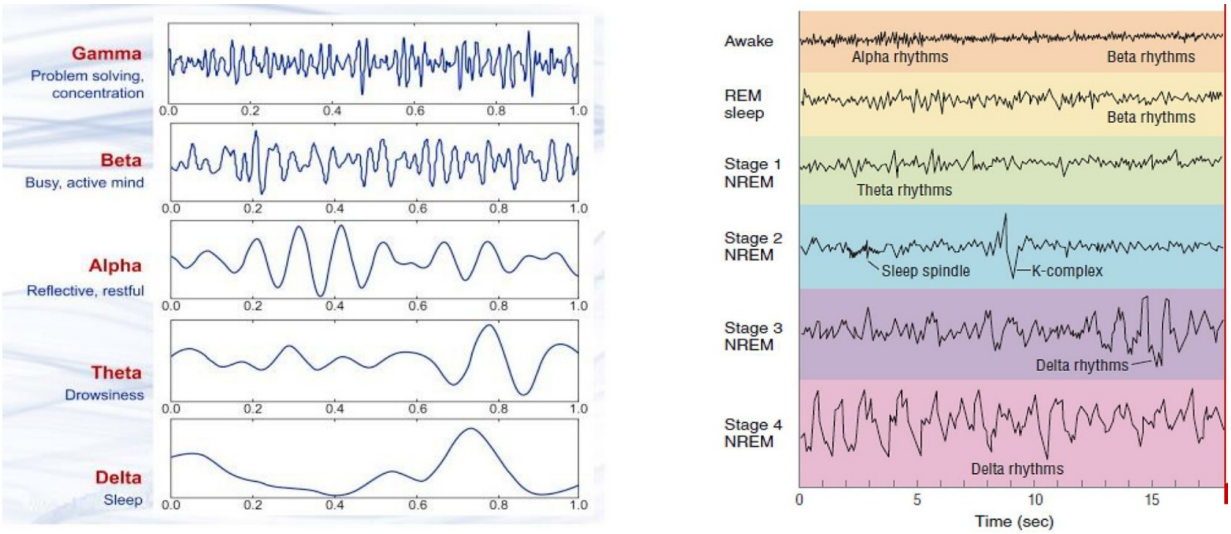
- Earlier attention focused mainly on diencephalon and anterior midbrain. Stimulation of medial bulbar RF and pontine / midbrain tegmentum helped reveal a brainstem arousal system, with thalamic projections contributing to cortical effects.
- The historical review contrasts these findings with earlier proposed seats of the mind or awareness, including ventricles, spinal cord, pineal gland, corpus striatum and periaqueductal gray.

25 Electroencephalography and brain rhythms

• **Electroencephalography (EEG)** records brain electrical activity non-invasively using scalp electrodes. The measured voltage differences arise mainly from summed synchronised postsynaptic currents in neuronal populations.



Scalp EEG recording



EEG rhythms and sleep-stage waveforms

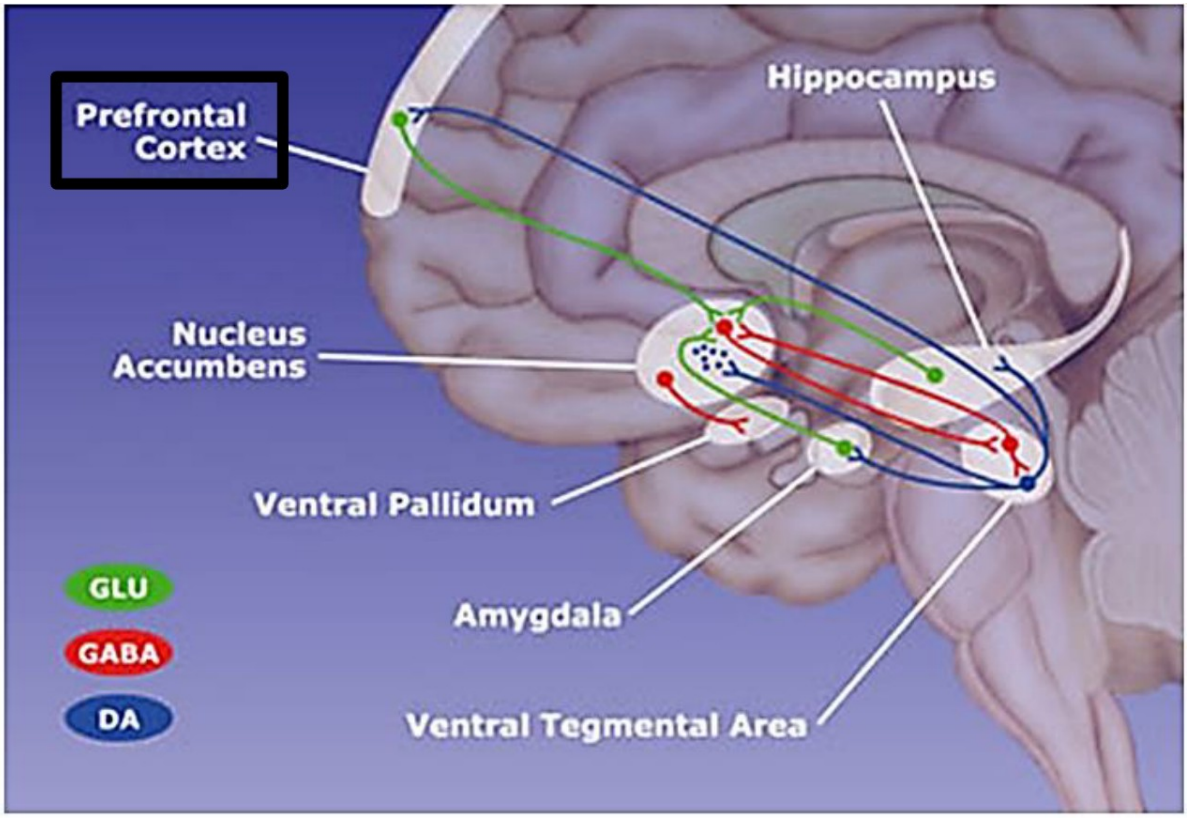
Pattern	Typical association illustrated
Gamma / beta	Gamma: concentration and problem solving; beta: active mental activity. These are associations, not unique mental-state signatures.
Alpha / theta	Alpha: relaxed wakefulness; theta: drowsiness and light sleep.
Delta	Slow-wave, deep non-rapid eye movement sleep.
Sleep-stage features	Stage 2: sleep spindles and K-complexes. Rapid eye movement (REM) sleep: relatively fast, low-amplitude activity.

• **Non-rapid eye movement (NREM)** stages 3 and 4 in the older diagram are combined as N3 in current scoring. EEG patterns vary with state; no single waveform identifies every case.

• EEG is widely used to evaluate **epilepsy** and conditions associated with **altered mental status or consciousness**.

26 Cortico-limbic reward circuitry

• The **mesocorticolimbic system** is a reward-related network comprising mesocortical and mesolimbic pathways. It connects motivation and reinforcement with cognitive control.



Mesocorticolimbic circuitry: structures and transmitter pathways

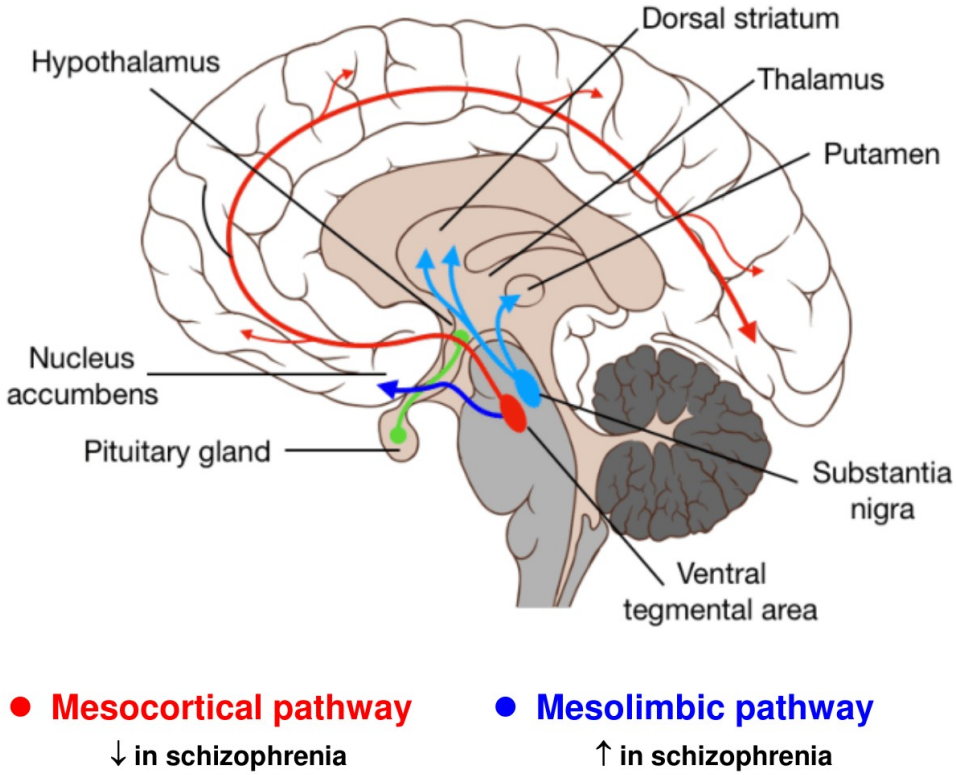
• The **prefrontal cortex (PFC)** is directly connected to limbic circuitry and contributes to **working memory, planning, decision-making, behavioural regulation and social cognition**.

Signal / node	Role in reading the circuit
Ventral tegmental area (VTA)	A midbrain source of dopamine (DA) projections to cortical and limbic targets.
Glutamate (GLU; green)	Predominantly excitatory projections, including cortical and limbic inputs to reward circuitry.
GABA (red)	Predominantly inhibitory projections; accumbens, ventral pallidal and midbrain interactions regulate downstream activity.
Dopamine (blue)	Modulates target processing. Its effect depends on receptor and circuit; it is not universally excitatory.

• Memory and emotional context interact with valuation and cortical control to shape motivated behaviour. The network contains reciprocal influences; it is not simply a one-way “pleasure pathway”.

27 Mesocortical versus mesolimbic pathways

Ventral tegmental area → **prefrontal cortex: mesocortical**
Ventral tegmental area → **limbic targets: mesolimbic**



Mesocortical and mesolimbic pathways

Pathway	Projection / significance
Mesocortical	VTA dopamine neurons → PFC; contributes to cognitive and executive regulation. The classical schizophrenia model associates reduced cortical dopamine function with cognitive / negative symptoms.
Mesolimbic	VTA dopamine neurons → limbic targets, especially nucleus accumbens; reward and motivation. The classical model associates increased mesolimbic dopamine function with positive symptoms.

Mesocortical ↓ ≠ mesolimbic ↑ in the classical dopamine account of schizophrenia

- These are teaching-model associations, not a complete explanation or a diagnostic test. Dopamine abnormalities and schizophrenia involve more complex, heterogeneous circuits.
- The wider diagram also labels dorsal striatum, putamen, substantia nigra, hypothalamus, thalamus and pituitary. Do not assign every coloured route to the VTA–accumbens pathway; other dopamine systems have different origins and targets.

28 Integrated revision • 1 of 2

Structures, processing and behaviour

Structure	Function / distinction to remember
Cerebral cortex	Outermost cerebral gray matter; sensory, motor and association functions.
White matter / basal nuclei	Myelinated connections / deep gray matter for movement and motivation.
Frontal lobe	Judgment, planning, personality, concentration, speech and movement.
Parietal / occipital	Somatosensation and multisensory integration / vision.
Temporal / insula	Hearing, smell, language, declarative memory, visual recognition / taste, vestibular and visceral sensation, motor integration.
Precentral / postcentral gyri	**Primary motor / primary somatosensory cortex**; separated by central sulcus.
Broca's / Wernicke's	**Motor speech / comprehension and word meaning**.
Association cortex	Recognises sensory patterns and combines information into meaning and thoughts.
Corpus callosum	Communication between hemispheres; most people have left language dominance.
RF / ARAS	Brainstem network / ascending component supporting cortical activation and arousal.
PFC / mesocorticolimbic system	Executive functions / interaction of cortical control with reward and motivation.

Three core pathways

Stimulus → receptor → sensory neuron → CNS integration → motor neuron → effector → response

Motor cortex → internal capsule → cerebral peduncle → medullary pyramid → **crossing** → lateral corticospinal tract → spinal motor circuitry → muscle

Sensory impulses → RF / ARAS → thalamic and cortical modulation → **alertness and attention**

- Homunculus size reflects sensory resolution or motor control demands. The map is topographic but not an exact fixed one-muscle-per-point layout.

29 Integrated revision • 2 of 2

Distinctions, dysfunction and causal reasoning

Comparison / change	Meaning or consequence
Sensation vs association	Detect a stimulus vs recognise, evaluate and interpret it.
Lateralisation vs isolation	Relative hemispheric specialisation vs the incorrect idea that complex skills belong entirely to one side.
Contralateral control	Most limb corticospinal fibres cross in the medulla; side of weakness depends on lesion level.
RF input vs output	Heterogeneous, multimodal inputs vs widespread, diffuse outputs.
Reduced wake-promoting drive	Can favour drowsiness / sleep; sleep is actively regulated and remains arousable.
RF / ARAS injury	Reduced cortical activation → impaired consciousness → possible coma. Severe associated brainstem injury can be fatal.
Habituation vs coma	Selective reduced response to familiar input vs pathological failure to arouse.
EEG	Scalp voltage recording; used in epilepsy and altered mental status / consciousness.
Fast vs slow EEG patterns	Fast low-amplitude activity often accompanies activation; delta is prominent in deep NREM sleep. REM shows why “sleep = slow EEG” is incomplete.
Mesocortical vs mesolimbic	VTA → PFC vs VTA → limbic targets, especially accumbens. Classical schizophrenia model: cortical ↓, limbic ↑.

Mechanisms to explain aloud

- Why does an alarm wake someone who sleeps through traffic? Familiar repeated input is habituated; a meaningful change can recruit arousal pathways.
- Why are fingers and lips large in the homunculus? Their high sensory resolution and fine motor demands require disproportionate cortical processing.
- Why can brainstem injury impair consciousness? Ascending activation of thalamic and cortical networks is disrupted; stronger injury can prevent arousal.
- How does reward circuitry relate to cognition? Dopamine-related limbic signals interact with memory, emotion and PFC functions to influence planning, decisions and behavioural regulation.